Scope of Work — Activities Page (Mbiufun Platform)

The Activities Page will be transformed from a simple content feed into a full **event creation, event participation, and social interaction system**. This Scope of Work outlines all required functional, frontend, backend, and integration responsibilities.

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1. Overview / Purpose

The Activities Page will allow users to:

- Create real-world activities or events
- Join existing activities
- View activity details (date, location, description, participants)
- Interact through MbiuHorn likes, comments, and shares
- See upcoming events in a clean, user-friendly interface

This page becomes the **action hub** of Mbiufun — connecting users through real-world plans.

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2. Core Features

The Activities Page will include the following main components:

A. Create Activity / Event

Users can create an event with:

- Title
- Description
- Date
- Start & end time
- Location
- Hobby category
- Optional image
- Privacy: Public / Friends Only / Invite Only
- Maximum participants

B. Join Activity

- Join button
- Cancel join
- Updated participant count
- Visual confirmation

C. Activity Details

- Full event info
- List of participants
- Activity creator
- Time & location display
- Optional map preview (if implemented later)

D. Engage with Activity

- MbiuHorn (like)
- Comment system
- Share (Facebook, Instagram, link)

E. Listing of Activities

- “Upcoming Activities” feed
- Sorted by date
- Show distance / relevance (optional future upgrade)

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3. Frontend Scope (Angular)

Components to Build

1. **activities-list.component**
2. **activities-create.component**
3. **activities-details.component**
4. **activities-join.component**
5. **activities-comments.component**
6. **mbiuhorn-like.component**
7. **activities-share.component**
8. **activities-card.component**

Frontend Tasks (Step-by-Step)

A. Activities List

1. Build list layout for all upcoming activities
2. Implement scroll & pagination
3. Display date, location, participants, hobby type
4. Add “Join / Joined” button
5. Add horn, comment count, share icons

B. Activity Creation Form

1. Form UI: title, description, date, time, location
2. Add upload image field
3. Add hobby picker
4. Add privacy selector
5. Add form validation & error messages
6. Build success modal

C. Activity Details Screen

1. Clean layout with hero banner
2. Event info: date, description, location
3. Participants list view
4. Comments section
5. Join / Cancel button state management
6. Share button popup

D. Horn & Comment Interactions

1. Horn animation on click
2. Comments UI with timestamp + user
3. Reply (optional)
4. UI logic for preventing spam

E. Share Integration

1. Create share modal
2. Add OS-native share API
3. Add fallback for copy-link

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4. Backend Scope (Django)

Backend Models

1. **Activity**
2. **ActivityParticipant**
3. **ActivityComment**
4. **ActivityHornBlast**

Backend Tasks (Step-by-Step)

A. Activity Management

1. Create Activity model (fields: title, description, date, time, location, hobby, capacity, etc.)

2. Create Activity CRUD endpoints
3. Validate date >= today
4. Validate capacity

B. Joining System

1. Create participants table
2. API: join event
3. API: cancel join
4. Prevent duplicate joins
5. Enforce max participants

C. Interaction System

1. MbiuHorn (like) endpoint
2. Prevent duplicate horns
3. Create comment API
4. Comment validation + sanitization

D. Activity Feed

1. Paginated endpoint for all upcoming events
2. Sorting by date/time
3. Filtering by hobby (optional)

E. Sharing Endpoint

1. Generate shareable event link
2. Ensure privacy rules
3. Optionally return preview data

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5. Expected Deliverables

- Fully functional Activities page
- All frontend components implemented and responsive
- Complete backend API layer with authentication
- Connected frontend & backend with clean event flows
- Validated, error-handled forms and interactions
- Share-ready activity links

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6. Dependencies / Integrations

- Django authentication & user model
- Angular routing & state management
- Optional: Google Maps API for locations
- OS-native sharing
- Database: PostgreSQL
